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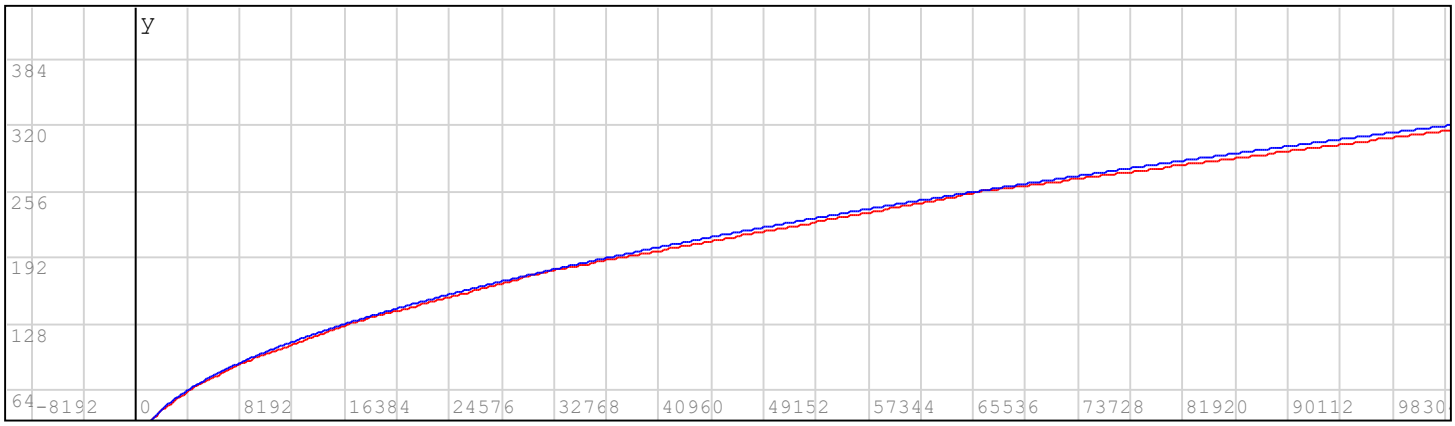
$$y(x) := \sqrt{x}$$

$$clz(x) := \begin{cases} 32 & \text{if } x = 0 \\ 32 - \operatorname{trunc}\left(\log_2\left(\operatorname{trunc}(x) + \frac{2^{32}-1}{2^{32}}\right) + \frac{2^{32}-1}{2^{32}}\right) & \text{else} \end{cases}$$

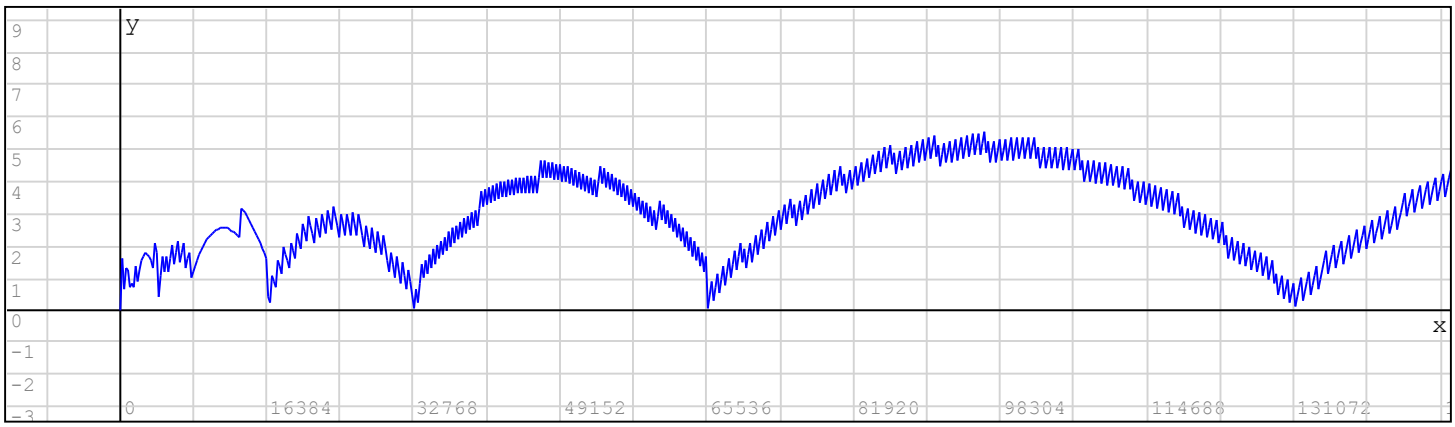
$$msb(x) := 32 - clz(x)$$

$$linScale(x) := \frac{\operatorname{trunc}\left(\operatorname{trunc}\left(\left(\sqrt{2}\right)^{msb(x)} - \left(\sqrt{2}\right)^{msb(x)-1}\right) \cdot 2^{32}\right)}{2^{32}}$$

$$linSqrt(x) := \operatorname{trunc}\left(\left(\frac{\left(\operatorname{trunc}(x) - 2^{msb(x)-1}\right)}{2^{msb(x)-1}}\right) \cdot linScale(x)\right) + \operatorname{trunc}\left(\sqrt{2^{msb(x)-1}}\right)$$



$\sqrt{x}$
$linSqrt(x)$



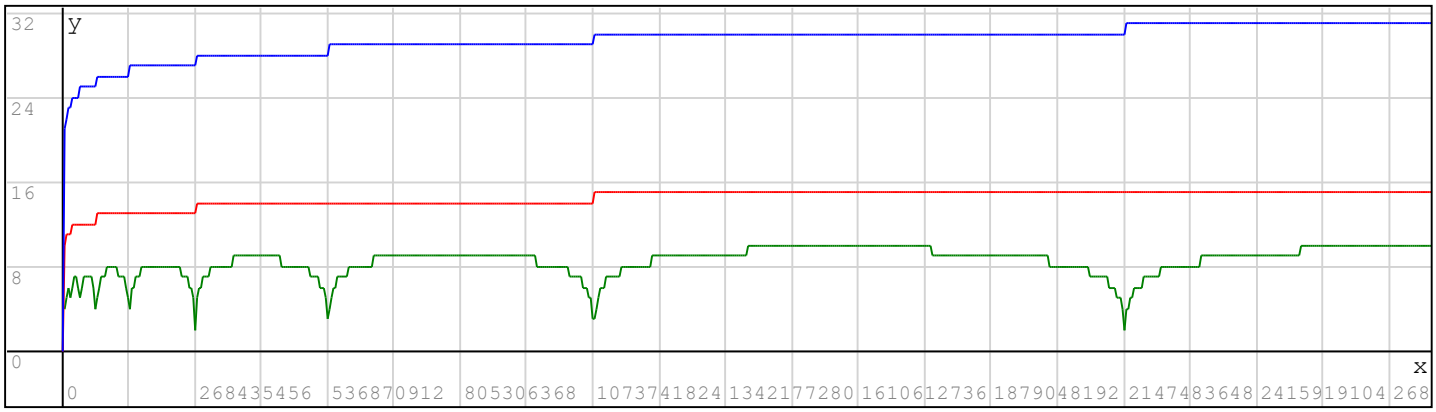
$\sqrt{x} - linSqrt(x)$

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$$argBitCount(x) := \text{trunc}\left(\log_2\left(x + \frac{2^{32}-1}{2^{32}}\right)\right) \qquad resultBitCount(x) := \text{trunc}\left(\log_2\left(\sqrt{\text{trunc}(x)} + \frac{2^{32}-1}{2^{32}}\right)\right)$$

$$linearBitError(x) := \text{trunc}\left(\log_2\left(\left(\sqrt{\text{trunc}(x)} - linSqrt(x)\right)\right) + \frac{2^{32}-1}{2^{32}}\right)$$



$$\begin{cases} argBitCount(x) \\ resultBitCount(x) \\ linearBitError(x) \end{cases}$$

$$linSqrtErr(x) := \sqrt{x} - linSqrt(x)$$

$$Max32 := \left(2^{32} - 1\right) \qquad PI4 := \left(\frac{2 \cdot \pi}{2^3}\right)$$

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$$Xcorr(x) := \mathrm{trunc}\left(\left(\frac{\mathrm{trunc}(x) - 2^{msb(x)-1}}{2^{msb(x)-1}}\right)\cdot\left(2^{32}-1\right)\right) \qquad QXcorr(x) := \mathrm{trunc}\left(\frac{Xcorr(x)\cdot Xcorr(x)}{2^{32}}\right)$$

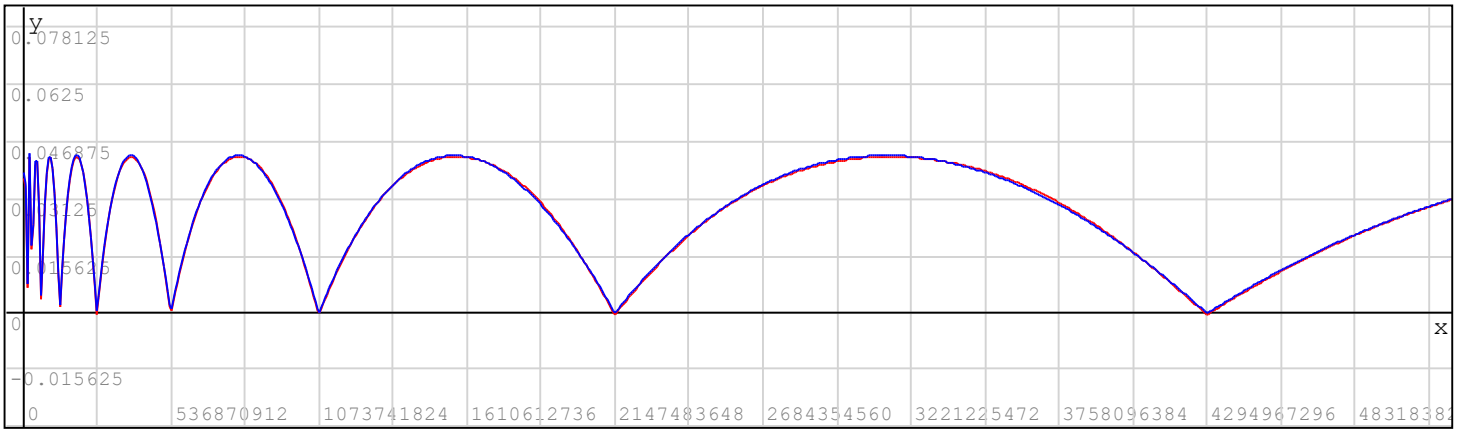
$$TXcorr(x) := \mathrm{trunc}\left(\frac{QXcorr(x)\cdot Xcorr(x)}{2^{32}}\right) \qquad FXcorr(x) := \mathrm{trunc}\left(\frac{TXcorr(x)\cdot Xcorr(x)}{2^{32}}\right)$$

$$QK := \mathrm{trunc}\left(Max32\cdot\frac{1.02035}{2}\right) \qquad TK := \mathrm{trunc}(Max32\cdot 0.480) \qquad FK := \mathrm{trunc}(Max32\cdot 0.52)$$

$$Xnew(x) := -Xcorr(x) + QXcorr(x)\cdot\frac{QK}{2^{31}} + TXcorr(x)\cdot\frac{(TK)}{2^{32}\cdot 2^{1}} - FXcorr(x)\cdot\frac{FK}{2^{32}\cdot 2^{1}} + Max32$$

$$ec(x) := \mathrm{trunc}\left(\frac{\mathrm{trunc}\left(\sin\left(\frac{\mathrm{trunc}(Xnew(x)\cdot PI4)}{2^{30}}\right)\cdot\mathrm{trunc}\left(\frac{16.48}{2^{5}}\cdot Max32\right)\right)}{2^{3}} - \left(2^{20}+2^{18}\right)\right)\cdot\frac{1}{2^{32}}$$

$$err(x) := \frac{linSqrtErr(x)}{linScale(x)}$$

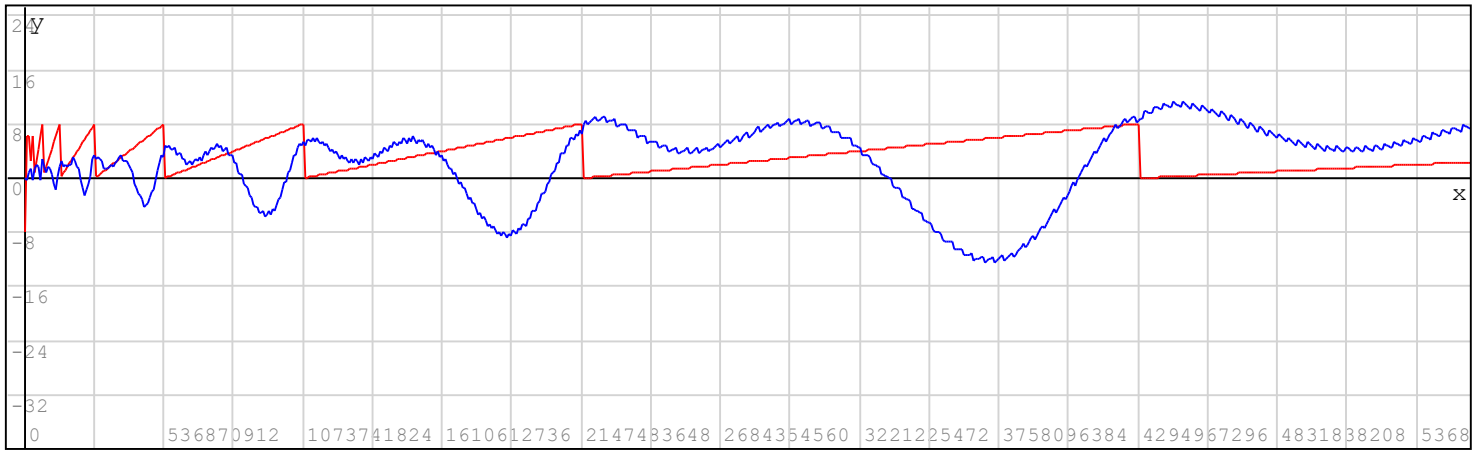


$$\begin{cases} err(x) \\ ec(x) \end{cases}$$

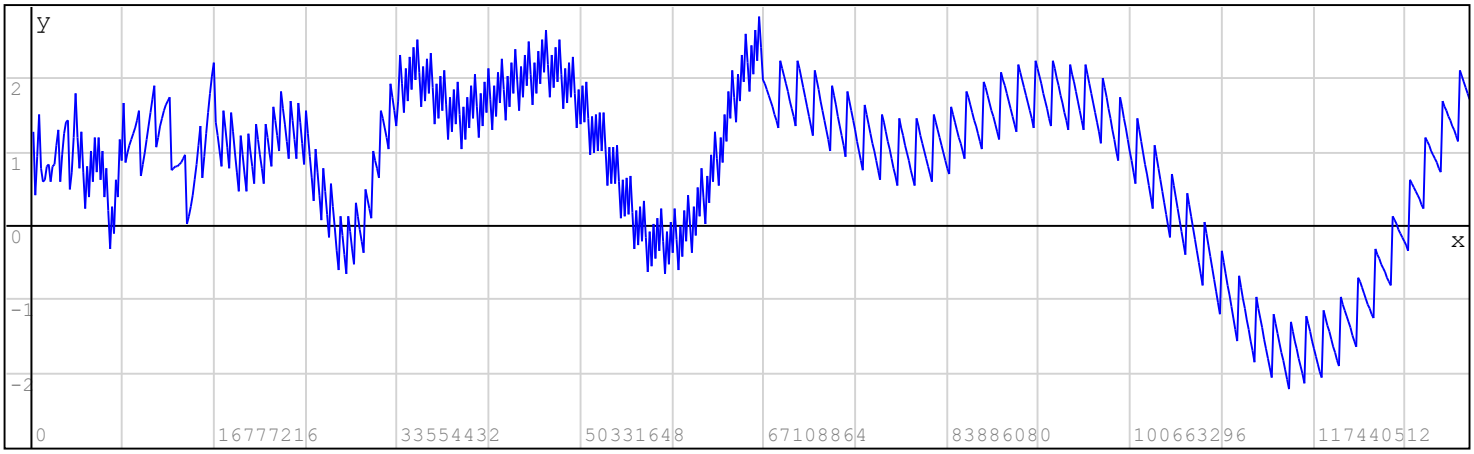
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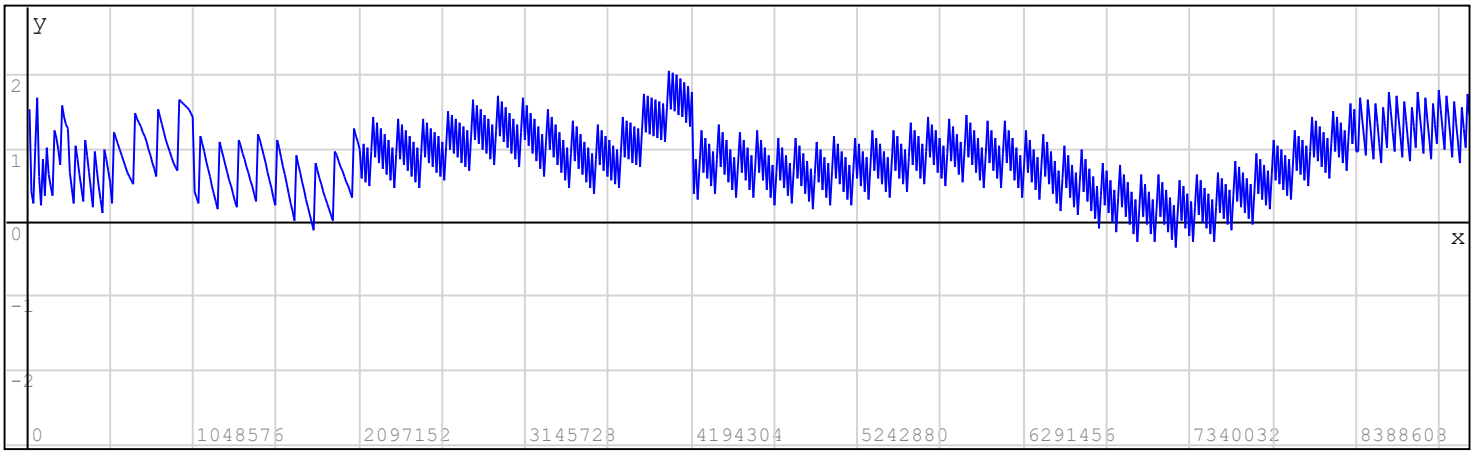
$cosSqrt(x) := linSqrt(x) + (ec(x) \cdot linScale(x))$



$$\begin{cases} \sqrt{x} - cosSqrt(x) \\ \dfrac{Xcorr(x)}{2^{29}} \end{cases}$$



$\left\{\sqrt{x} - cosSqrt(x)\right.$



$\left\{\sqrt{x} - cosSqrt(x)\right.$

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